

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Botany**Paper: BOB-501 — Comparative Studies of Cryptogams**

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, selecting at least one from each Sections A, B and C and Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Briefly answer the following in not more than 50 words each: [10 × 1 = 10]

- (a) Define coenobium.
- (b) Differentiate between zoospore and hypnospore.
- (c) What is the function of the Retort cell and its location?
- (d) Differentiate between isomorphic and heteromorphic alternation of generations.
- (e) What do you mean by perfect state of the fungi?
- (f) Name one fungus used in the pharmaceutical industry.
- (g) What is diplanetism? In which group of fungi is it observed?
- (h) Differentiate between ectophloic siphonostele and amphiphloic siphonostele.
- (i) Differentiate between true and false indusium.
- (j) Write the systematic position of *Psilotum*.

SECTION A

Question 2. Write a comparative account of various types of the thallus organization found in the algae. Describe the evolutionary tendency found in the order Volvocales of Chlorophyceae. [15]

Question 3. Discuss any two of the following: [2 × 7.5 = 15]

- (a) Chromatic Acclimation (CA) in Cyanobacteria
- (b) Sexual reproduction in members of Oedogoniales
- (c) Trimorphic life history

Question 4. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Life history of *Ectocarpus* sp.

- (b) Photosynthetic pigments in algae and the Cyanobacteria
- (c) T.S. thallus of *Codium* sp.

SECTION B

Question 5. Give an account of the Ainsworth's system of classification of the fungi. [15]

Question 6. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Various types of fungal spores
- (b) Types of life cycles in the fungi
- (c) Economic importance of the fungi

Question 7. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Liberation of the spores in the fungi
- (b) General characteristics of Basidiomycotina
- (c) Mode of sexual reproduction in the fungi

SECTION C

Question 8. Write a comparative account of sporophytes of *Sphagnum* and *Polytrichum* with labelled diagrams. [15]

Question 9. Describe the reproduction in *Marsilea* with the help of suitable diagrams. [15]

Question 10. Answer any two of the following: [2 × 7.5 = 15]

- (a) Write about the development of sporangium in *Isoetes*.
- (b) Draw labelled diagrams of T.S. of *Notothylas* gametophyte (description not required).
- (c) Draw labelled diagrams of T.S. of *Sphagnum* and *Polytrichum* leaf (description not required).

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